

Scalability & Future Plan

Date	15 July 2026
Team ID	*****
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	1 Mark

This document outlines how the current solution can scale and the roadmap for enhancing it beyond its current state.

Current System Limitations

Limitation	Impact	Priority to Address
No database integration for storing applicant records.	Submitted applications cannot be stored or reviewed later.	High
Model is trained on a public Loan Prediction dataset (614 records).	Prediction accuracy may differ from real banking data.	High
No user authentication or authorization.	Anyone can access the application and submit loan details.	Medium
Application runs only on a local Flask server.	Cannot be accessed remotely without deployment.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local Flask Development Server	Deploy using Gunicorn and Nginx or cloud platforms such as Render, AWS, or Azure to support more users.
2	Data Storage	No runtime database	Integrate MySQL , PostgreSQL , or SQLite to store applicant information and prediction history.

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2	Data Storage	No runtime database	Integrate MySQL , PostgreSQL , or SQLite to store applicant information and prediction history.

3	Performance	Fast prediction using a trained Random Forest model	Optimize application performance through cloud deployment and efficient request handling for multiple users.
4	Security	No authentication	Implement secure login, user authentication, and role-based access for administrators and loan officers.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Database integration for storing applicant records and prediction history	Within 2 months	Enables record management and future model retraining.
Phase 3	Deploy the application on a cloud platform (Render, AWS, or Azure)	Within 3 months	Makes the application accessible online with improved scalability.
Phase 4	Add user authentication and an administrator dashboard	Within 4 months	Improves security and allows application management by authorized users.
Phase 5	Retrain the model using real banking datasets and integrate explainable AI features	Within 6 months	Increases prediction accuracy, transparency, and reliability for real-world loan approval.